Dr Antonis (Antony) Vamvakeros

Research Engineer / Applied Scientist — Machine Learning for Scientific Data

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Profile

My work focuses on the development and application of multimodal imaging to study functional materials and devices, like catalysts, fuel cells and batteries, under industrially relevant conditions (*in situ*/*operando* experiments). I am the developer of nDTomo, an open source library for handling chemical imaging data as well as deep learning approaches for ultra-fast analysis of large diffraction datasets, CT image reconstruction and CT artefact reduction. I am the creator of the Battery Imaging Library, the first open access multimodal multi-length scale imaging database.

Experience

Nov 2022 - present**: Royal Society Industry Fellow, Dyson School of Design Engineering, Imperial**

- Cutting-edge Research: Developed physics-informed and generative AI models (GANs, flow-matching) for unpaired multi-modal multi-resolution imaging data fusion
- Project Founder: Conceived and created the Battery Imaging Library (BIL), the first open-access multi-modal and multi-length-scale imaging database for battery research. I led an international team of 48 researchers from the UK, France, Germany, USA and Japan.
- Supervision: Mentoring a PhD student on image segmentation using Meta's SAM and Dino foundation models; steered work to successful presentation at NeurIPS 2024 AI4Mat Workshop

Feb 2021 - present: **Research & Development Lead Scientist, Finden Ltd**

- Team Management: Leading the scientific computing team consisting of three scientists
- Grant & PI Leadership: Technical lead and co-PI for £900k+ in UKRI funded R&D projects
- Software Development: Directed the engineering development of BeamStop, Finden's first commercial software for chemical imaging and tomography data
- Code IP & Licensing: Secured IP licensing for proprietary PQ-Net neural network code to an OEM and open-source nDTomo code integrated into the DECTRIS Ltd cloud platform

May 2017 - Feb 2021: **Research Scientist, Finden Ltd**

- Machine Learning for Industry: Developed computer vision and X-ray CT metal artefact reduction models for a UK government agency
- Experiments for Industry: Performed advanced materials characterisation for industrial clients
- Inhouse R&D: Designed and deployed ML methods for tomographic reconstruction, data denoising, artefact reduction, image segmentation and fast spectroscopic data analysis

Jan 2019 - Jan 2021**: Honorary Research Fellow, Dept. of Chemistry, UCL**

- Technical Breakthrough: Developed the Direct Least-Squares Reconstruction algorithm, providing the first solution to the parallax problem in scatter-based CT

May 2017 - Jan 2019: **Postdoctoral Scientist, European Synchrotron Radiation Facilities**

- Innovation: Developed 5D tomographic diffraction imaging and deployed real-time hyperspectral CT reconstruction and analysis tools for facility users during live experiments

Education

2017: **PhD in Dynamic Hyperspectral Tomography (Chemistry)**, University College London, UK
2013: **MSc in Chemical Process Engineering**, University College London (Distinction), UK
2012: **MEng in Chemical Engineering**, University of Patras, Greece

Selected Open-Source Code and Data: nDTomo, BIL, Vulture, HR-Dv2, SAMBA, SD2I

Awards

- Royal Society of Chemistry 2023 Analytical Science Horizon Prize: "The XRD-CT Pioneers"
- Clark Prize, Department of Chemistry, UCL (2016)